

Project Proposal: VoiceArmor-Edge

Real-Time On-Device Voice Spoof & Synthetic Call Detection
for Snapdragon-Powered HP PCs

Challenge: Snapdragon AI Lab Build & Present Challenge 2026

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Discipline: Electronics & Communication Engineering (ECE)

Target Hardware: Snapdragon X-powered HP OmniBook PCs (Hexagon NPU) + Arduino UNO Q

1. Project Overview

AI tools can now clone a person's voice using just a short 3-second audio recording. Scammers are using these cloned voices to impersonate police officers, bank officials or family members to steal money -- a major issue right now with "Digital Arrest" scams in India.

Current security solutions rely on cloud servers to analyze audio. This does not work for live calls:

- Sending call audio over the internet is too slow to warn someone in real time.

- It req

VoiceArmor-Edge solves this problem by running entirely on the laptop using the Snapdragon Hexagon NPU. It listens to incoming call audio in short 250ms chunks, converts the sound into frequency pictures (spectrograms) and runs a lightweight AI model to check if the voice is real or fake in under 15ms.

To give users a physical warning they cannot miss, VoiceArmor-Edge also connects via USB to an Arduino UNO Q board. If a fake voice is detected, the Arduino immediately turns on a bright red warning light and buzzer.

2. Real-World Need & Use Case

The Problem

- **Voice Cloning Fraud:** Fraudsters call victims pretending to be law enforcement or family members in distress, demanding urgent bank transfers.

- No P

Why It Must Run on the Snapdragon NPU

- **Fast Response:** Checking speech on the NPU takes less than 15ms, warning the user while the caller is still speaking.

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3. Technical Implementation

A. Audio Feature Extraction (Digital Signal Processing)

When AI tools generate fake speech, they leave subtle irregularities in higher frequencies that natural human vocal cords do not produce.

- The application captures audio from your computer's audio output in real time.

- It slices the audio into small segments (e.g., 100ms) and represents each segment as a vector.

B. Qualcomm AI Hub Model

- The classifier is adapted from standard audio models in the Qualcomm AI Hub catalog.

- INT8 quantization is used for inference. This significantly reduces memory footprint and improves inference speed.

C. Running on Windows on Snapdragon

- The app runs as a native Windows service using `onnxruntime` with the `onnxruntime-qnn` execution provider plugin.

- The model is loaded directly from the Qualcomm AI Hub, ensuring compatibility and optimal performance.

D. Hardware Alert with Arduino UNO Q

In line with the Snapdragon AI Lab setup:

- The Snapdragon HP PC does the heavy AI computation.

- The Arduino Uno Q provides a hardware alert (e.g., LED, buzzer) when a high-confidence detection is made.

4. Why This Approach Stands Out

- **Practical & Focused:** Instead of trying to do too many things at once, it solves one critical, high-impact problem very well -- live synthetic voice detection.

- **True**

5. Deployment on Snapdragon HP PCs

- **Target Device:** HP OmniBook laptops powered by Snapdragon X Elite / X Plus.

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6. Project Presentation & Code

- **Slide Deck:** A clean 10-slide presentation explaining the problem, architecture, NPU integration and future roadmap.

- **Sourc**

7. Conclusion

VoiceArmor-Edge is a focused, realistic and practical project that shows the real value of the Snapdragon Hexagon NPU. By combining digital signal processing, Qualcomm AI Hub model optimization and a physical hardware alert, it protects everyday users from growing AI voice scams safely and privately on their own laptops.